GWAS Collaboration

Workflow at a Glance

Initiator • Collaboration Server • Collaborator

Short version — 7 key steps

Companion to the full timeline. Login/authentication and backend internals are omitted.

How to read this document

Each numbered step below is one stage of a GWAS collaboration, from registering cohorts to reading the final AI-generated summary report.

Every step shows three columns: what the Initiator (study lead) is doing, what the Collaboration Server is doing, and what the Collaborator is doing — at the same point in time.

A purple ▸ ON MACHINE banner means that step's work happens on that party's own computer. Raw genotype data never leaves it — only counts, survivor lists, and summary statistics are returned.

Each step has 1–3 screenshot placeholders below the swimlane — drop in a screenshot of the actual UI for that stage.

The 7 steps

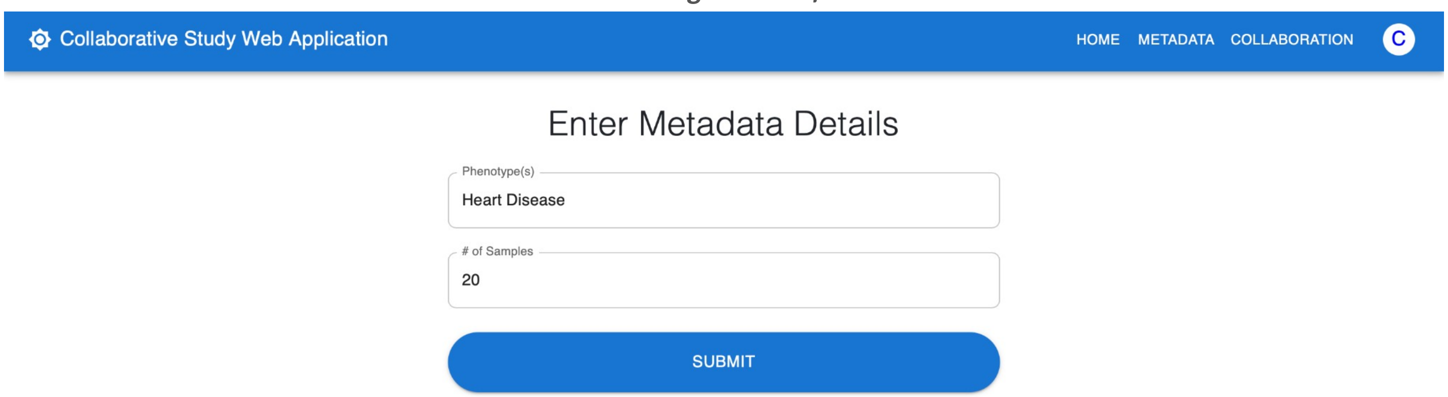
Step	Stage	What happens
1	**Register the cohort**	*Each party tells the platform about its data*
2	**Set up the collaboration**	*Initiator configures the study and invites partners*
3	**Accept the invitation**	*Collaborator agrees to participate*
4	**Quality Control**	*Filter samples and SNPs — privately on each machine*
5	**Generate summary statistics**	*Turn QC-filtered samples into per-SNP counts*
6	**Run GWAS analysis**	*Combine counts across parties and view results*
7	**AI summary report**	*Plain-English, privacy-preserving overview*

Step 1 • Register the cohort

Each party tells the platform about its data

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
1 *Each party tells the platform about its data*	▸ ON INITIATOR'S MACHINE • Logs in and adds a record describing their cohort: phenotype (e.g. eye colour, heart disease) and sample count. • Their raw genotype file stays on their own computer.	• Stores only the cohort description so other researchers can discover it later. • Never receives the underlying genotype data.	▸ ON COLLABORATOR'S MACHINE • Does the same on their own computer — registers their cohort with phenotype + sample count. • Their raw data also stays local.

Cohort registration form

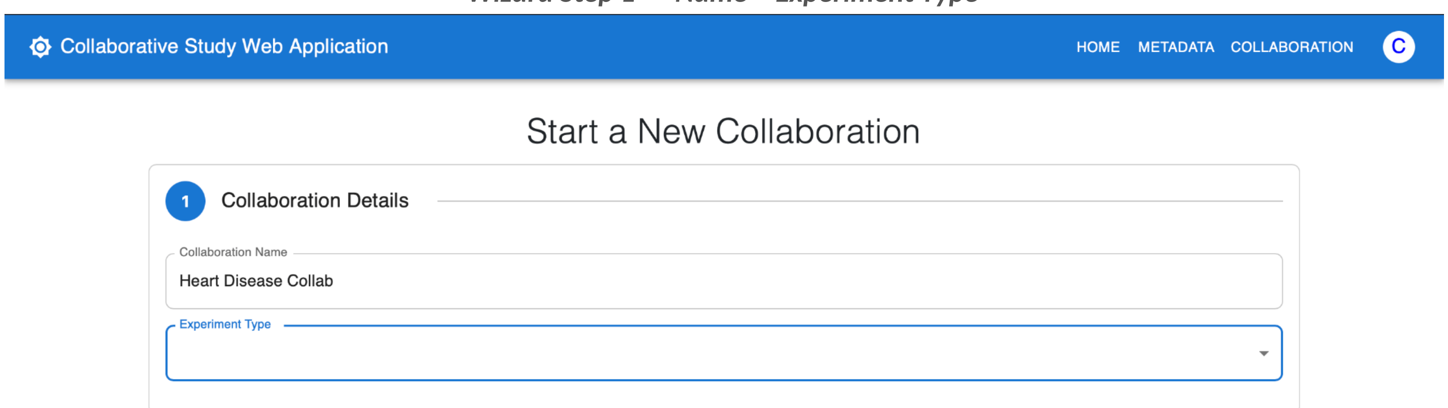


Step 2 • Set up the collaboration

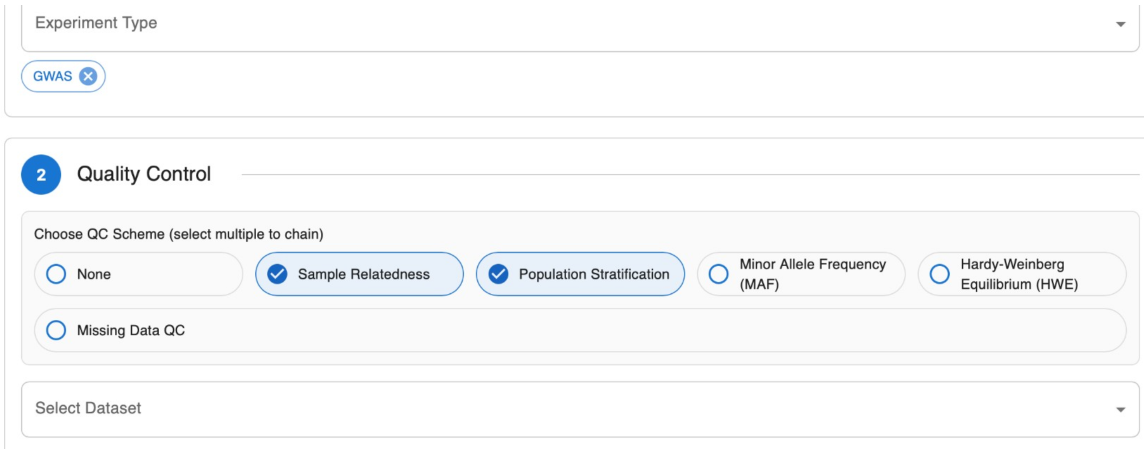
Initiator configures the study and invites partners

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
2 *Initiator configures the study and invites partners*	• Opens the Collaborations page and clicks Start a Collaboration. • STEP 1: gives the collaboration a name and picks Experiment Type = GWAS. • STEP 2: picks one or more Quality Control methods (and sets threshold values for methods that need them). • Selects which dataset to use for the study. • STEP 3: searches for partners by phenotype and sample size, then sends invitations and submits.	• Creates the collaboration record with all the chosen settings. • Sends invitations to each selected researcher.	• Receives a notification that they have been invited.

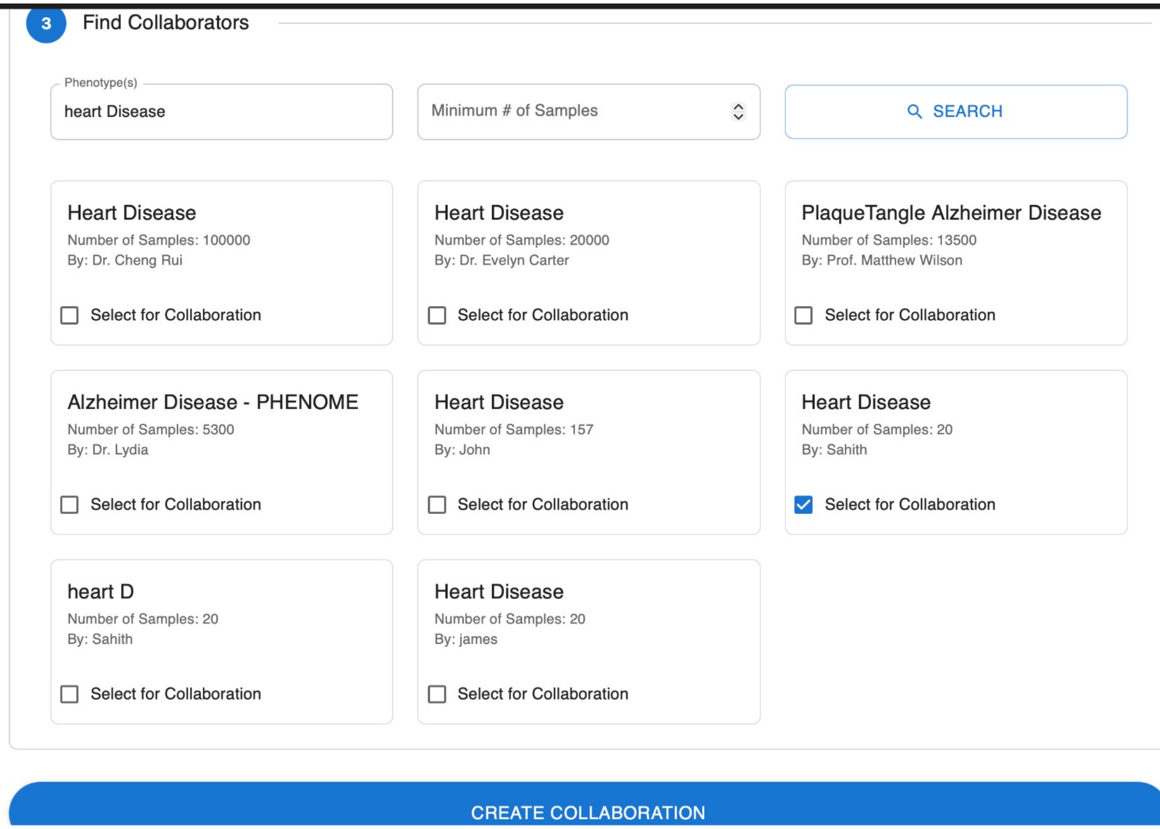
Wizard Step 1 — Name + Experiment Type



Wizard Step 2 — QC methods + thresholds



Wizard Step 3 — Find Collaborators + Submit

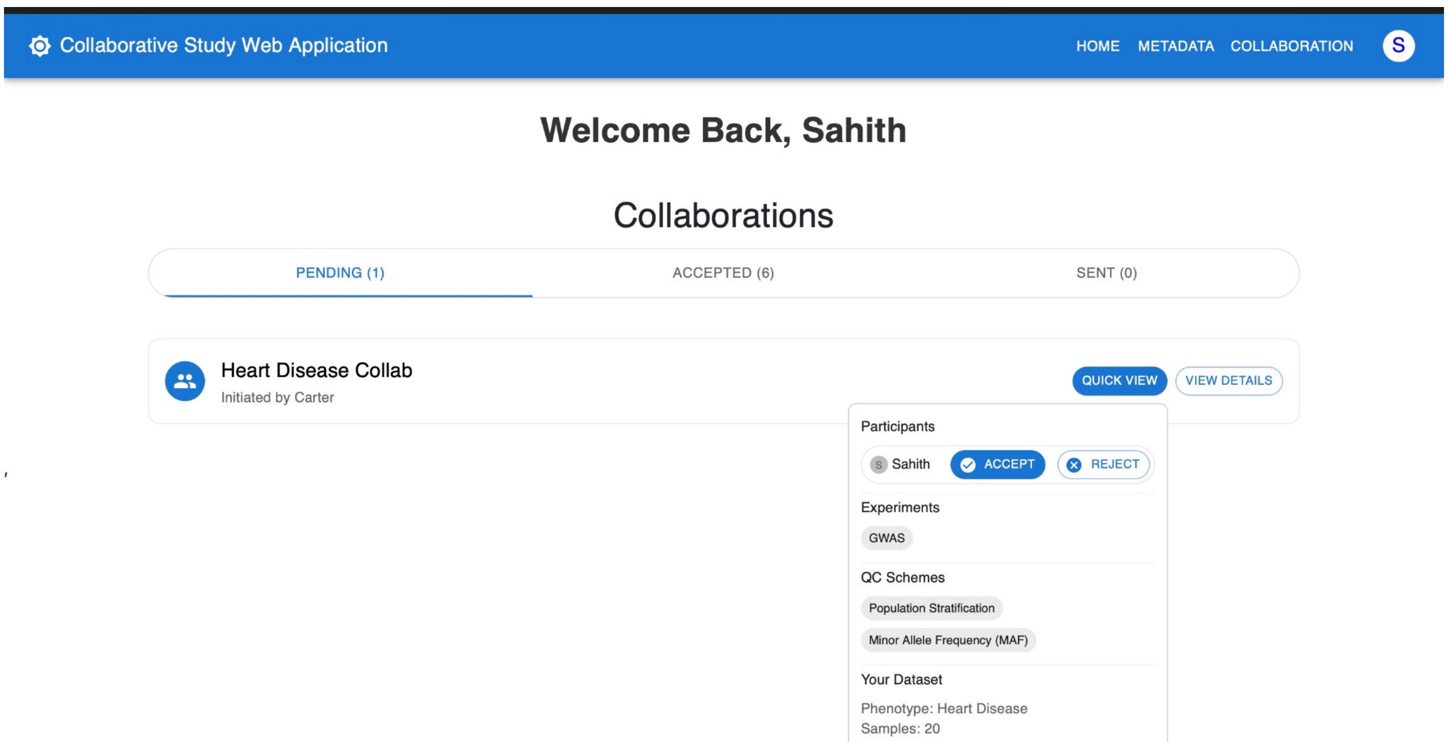

3
Find Collaborators
Phenotype(s)
heart Disease
Minimum # of Samples
SEARCH
Heart Disease
Number of Samples: 100000
By: Dr. Cheng Rui
Select for Collaboration
Heart Disease
Number of Samples: 20000
By: Dr. Evelyn Carter
Select for Collaboration
PlaqueTangle Alzheimer Disease
Number of Samples: 13500
By: Prof. Matthew Wilson
Select for Collaboration
Alzheimer Disease - PHENOME
Number of Samples: 5300
By: Dr. Lydia
Select for Collaboration
Heart Disease
Number of Samples: 157
By: John
Select for Collaboration
Heart Disease
Number of Samples: 20
By: Sahith
Select for Collaboration
heart D
Number of Samples: 20
By: Sahith
Select for Collaboration
Heart Disease
Number of Samples: 20
By: james
Select for Collaboration
CREATE COLLABORATION

Step 3 • Accept the invitation

Collaborator agrees to participate

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
3 *Collaborator agrees to participate*	• Watches the collaboration page — the invited researcher's status changes from pending to accepted.	• Records the acceptance and is ready to start Quality Control.	• Reviews the invitation (study name, QC plan, phenotype) and clicks Accept.

Collaborator's invitation card with Accept / Reject

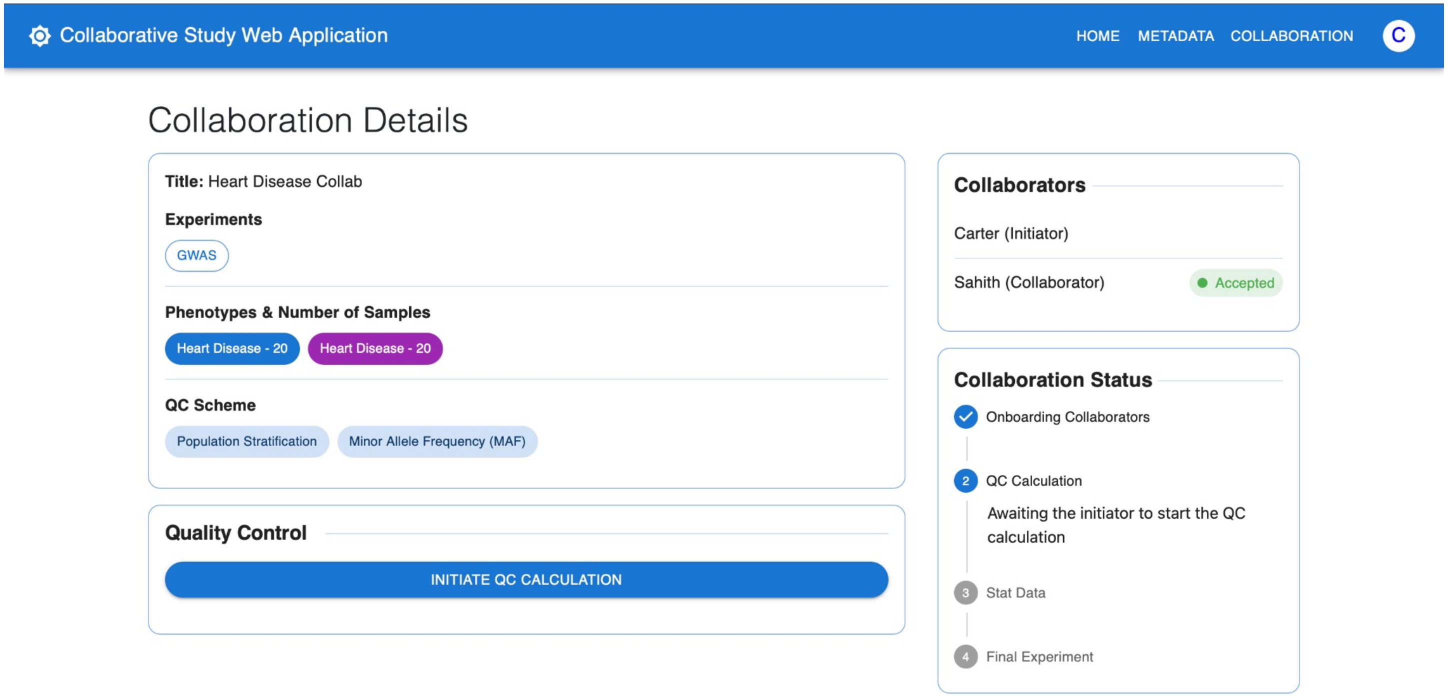


Step 4 • Quality Control

Filter samples and SNPs — privately on each machine

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
4 *Filter samples and SNPs — privately on each machine*	▸ ON INITIATOR'S MACHINE • Filter-only QC (MAF / HWE / Missing Data) runs automatically — survivor counts show up on its own. • For pairwise methods (Sample Relatedness, Population Stratification) clicks Initiate QC Calculation. • Drags a Threshold slider — the per-party sample counts update LIVE as the slider moves. • Clicks Confirm Threshold to lock in the surviving samples and reveal the per-party preview list.	• Orchestrates QC across all parties. • Receives only the survivor lists — never sees raw genotype data.	▸ ON COLLABORATOR'S MACHINE • QC runs on their own machine against their raw data. Raw data never leaves their computer. • Sees the same live counts and the same preview list once the initiator confirms.

QC running on each party's machine (per-method progress)



[SCREENSHOT]

Initiate QC Calculation + threshold slider + preview list

For pairwise methods: Initiate button, slider with live per-party counts, Confirm Threshold, preview list.

Step 5 • Generate summary statistics

Turn QC-filtered samples into per-SNP counts

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
5 *Turn QC-filtered samples into per-SNP counts*	▸ ON INITIATOR'S MACHINE • Clicks Create GWAS Dataset. • Per-SNP case/control counts are computed on their own machine and only those counts are sent to the server.	• Collects each party's aggregated counts. • Marks the collaboration as ready for analysis once everyone has uploaded.	▸ ON COLLABORATOR'S MACHINE • Does the same on their machine — counts only, never raw data.

[SCREENSHOT]

Stat upload status — each party Uploaded

Per-party checklist showing all rows flipped to Uploaded.

Step 6 • Run GWAS analysis

Combine counts across parties and view results

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
6 *Combine counts across parties and view results*	• Clicks Initiate GWAS Calculation. • Sees a ranked table of SNPs with chi-square and p-value, sorted by significance. • Can export the results.	• Combines the per-SNP counts from all parties. • Computes the chi-square test for each SNP and stores the results.	• Sees the same results table — identified only by site label, not by name.

[SCREENSHOT]

GWAS results table (top SNPs)

Table sorted by p-value / chi-square.

Step 7 • AI summary report

Plain-English, privacy-preserving overview

STEP	INITIATOR	COLLABORATION SERVER	COLLABORATOR
7 *Plain-English, privacy-preserving overview*	• Clicks Generate Summary. • Reads the AI-written report: overview, top SNPs, per-site contribution, and a recommendation (continue / continue with caveats / reconsider).	• Builds an anonymised digest (collaborators shown as Site A, Site B, …) and sends it to the AI model. • Stores the generated report so subsequent reads are instant.	• Opens the collaboration and reads the same report. • Identified only by site label — never by real name or institution.

[SCREENSHOT]

Final AI Summary card with recommendation banner

Overview + top SNPs + per-site analysis + recommendation callout.

Privacy at a glance

- Raw genotype data never leaves the machine it sits on. Quality Control runs locally on each party's computer; only survivor lists are returned to the Collaboration Server.
- The GWAS step uses aggregated case/control counts — there is no row-level genotype exchange between parties.
- The AI summary is built from an already-aggregated, anonymised digest. Collaborators are shown as Site A / Site B / … in the report.